

Introduction

This report presents a Sudoku Solver developed in C++. The program solves a 9x9 Sudoku puzzle using the backtracking algorithm. It reads a partially filled grid and fills in the empty cells with valid numbers.

Code Overview

isSafe() — Checks if a number can be placed in a given cell.

solveSudoku() — Recursively solves the puzzle using backtracking.

printGrid() — Displays the Sudoku grid.

Output

```

C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task3\sudoku_solver.exe
=====
CodeAlpha SUDOKU SOLVER
=====

Original Puzzle:
=====
          SUDOKU GRID
=====
5 3 0 | 0 7 0 | 0 0 0
6 0 0 | 1 9 5 | 0 0 0
0 9 8 | 0 0 0 | 0 6 0
-----
8 0 0 | 0 6 0 | 0 0 3
4 0 0 | 8 0 3 | 0 0 1
7 0 0 | 0 2 0 | 0 0 6
-----
0 6 0 | 0 0 0 | 2 8 0
0 0 0 | 4 1 9 | 0 0 5
0 0 0 | 0 8 0 | 0 7 9
=====

Solving...

[OK] Puzzle Solved!

=====
          SUDOKU GRID
=====
5 3 4 | 6 7 8 | 9 1 2
6 7 2 | 1 9 5 | 3 4 8
1 9 8 | 3 4 2 | 5 6 7
-----
8 5 9 | 7 6 1 | 4 2 3
4 2 6 | 8 5 3 | 7 9 1
7 1 3 | 9 2 4 | 8 5 6
-----
9 6 1 | 5 3 7 | 2 8 4
2 8 7 | 4 1 9 | 6 3 5
3 4 5 | 2 8 6 | 1 7 9
=====

Thank you for using CodeAlpha Sudoku Solver!
=====

-----
Process exited after 0.2666 seconds with return value 0
Press any key to continue . . .

```

Figure 1: Sudoku Solver Output

Features

2D Array Representation: The Sudoku grid is stored as a 9x9 2D array.

Backtracking Algorithm: The solver tries numbers 1-9 in empty cells and backtracks when a conflict occurs.

Rule Checking: Validates row, column, and 3x3 subgrid constraints.

Formatted Output: Displays the puzzle and solution in a clean grid format.

Conclusion

The Sudoku Solver demonstrates the use of 2D arrays, recursion, and backtracking in C++. It efficiently solves any valid Sudoku puzzle.

References

C++ Documentation. (2026). Arrays and Recursion. Retrieved from <https://en.cppreference.com>

GeeksforGeeks. (2025). Sudoku Solver using Backtracking. Retrieved from <https://www.geeksforgeeks.org>